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1 March 1978
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MILITARY SPECIFICATION

PORK LOIN ROASTS AND SLICES, BONELESS, FROZEN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers frozen, boneless, pork loin roasts and slices for use by the Armed Forces in kitchens where refrigeration facilities are available.

1.2 Classification. The product shall be of the following types, as specified (see 6.1):

Types

I - Roasts
II - Slices

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Command, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8905

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SPECIFICATIONS

FEDERAL

L-P-375	-	Plastic Film, Flexible, Vinyl Chloride
L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin
PP-P-571	-	Pork; Fresh (Chilled or Frozen)
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-1163	-	Box, Corrugated Fiberboard, High Compression Strength, Water Resistant, Wax Resin Impregnated

MILITARY

MIL-M-1018	-	Meat Cuts and Sausage Products (Chilled or Frozen) for Domestic and Overseas Shipment, Packaging, Packing, Labeling and Marking Of
MIL-C-43619	-	Cloth, Netting, Cotton, Elasticized, Raschel Knit

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-668	-	Sanitary Standards for Food Plants

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

LAWS AND REGULATIONS

US DEPARTMENT OF HEALTH, EDUCATION AND WELFARE

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder

US DEPARTMENT OF AGRICULTURE

Meat and Poultry Inspection Regulations

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

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NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N. W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 380 - Standard for Metric Practice

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Pork loins. Bone-in pork loins shall comply with the following requirements:

1. Shall be fresh (not previously frozen) chilled bone-in, blade-in pork loins.

2. Shall be in excellent condition (see 6.3) at the time of offering or boning, and shall be free of foreign materials (see 4.3.1.1).

3. Internal temperature at the thickest part of the bone-in loins shall be in a temperature range of 28°F (-2°C) to 40°F (4°C) at any time after initial chilling following slaughter and prior to boning. Alternatively, the internal temperature at the thickest portion of the bone-in loin shall be in a temperature range of 28°F to 46°F (-2°C to 8°C) inclusive at any time after initial chilling, following slaughter and prior to boning and the bone-in loin shall be offered within 48 hours of the stated kill date as evidenced by a USDA, kill date certificate.

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4. Shall show no evidence of freezing, thawing or freezer-burn.

5. Shall weigh not more than 20 pounds (9.07 kg) plus a tolerance of 1/2 pound (0.23 kg) and contain not less than 12 ribs.

6. Shall not be trimmed of lean meat for the purpose of meeting weight requirements. The tenderloin shall remain intact in the loin to determine the bone-in weight.

7. The ham end shall contain not less than one complete nor more than four sacral vertebrae and shall terminate on a line which is reasonably straight and perpendicular to the split surface of the backbone.

8. The anterior surface of the blade end shall be reasonably straight and perpendicular to the split surface of the backbone.

9. The line of separation of the loin from the belly shall extend from a point on the first full rib (blade end) of the loin which shall measure not more than 2-1/4 inches (57 mm) from the junction of the first full rib and the foremost thoracic vertebrae to a point on the ham end which is immediately adjacent to the major tenderloin muscle. The line of separation shall be reasonably parallel to the major loin muscle.

3.1.1.1 General selection requirements. The pork loins shall comply with the following requirements:

1. Shall be thick, uniformly full and well rounded.

2. Cartilage shall be in evidence.

3. Split chine bones, spinous processes and cross-cut sections of bones shall be porous and not brittle or flinty.

4. Color of the bones shall range from red to deep pink.

5. Exterior surface of the rib bones shall show at least a slight red color.

6. The exterior fat shall be firm and white.

7. The lean shall be fine texture, and firm, showing at least slight to moderate amount of intramuscular fat (marbling) in the longissimus dorsi and spinalis muscles at the blade end.

8. The lean shall possess a bright and uniform color (slightly two-toned permissible) ranging from light pink or grayish pink to bright red.

9. The flesh shall not be dark, gummy, or oily.

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3.1.2 Binding agents. Powdered vital wheat gluten or other binding agents, providing satisfactory finished product (see 3.3), shall be used to bind together the lean interior surface of each half of the loin during processing prior to freezing (see 6.2). The binding agents shall be as permitted in the Meat and Poultry Inspection Regulations of the US Department of Agriculture and shall be free from foreign materials, off-flavors and off-odors (see 4.3.1.2).

3.2 Processing. The pork loins shall be so processed and handled that the types specified shall conform to the finished product requirements (see 3.3). In addition, the following process times and temperatures shall apply:

1. Internal temperature at the thickest part of the chilled bone-in pork loins prior to and at the start of boning. 28°F (-2°C) to 40°F (4°C) or 28°F (-2°C) to 46°F (8°C) as applicable (see 3.1.1)

2. Maximum internal temperature at the thickest part of the chilled boned pork loins after boning until the fabricated product is in the freezing process. 42°F (6°C) 1/

3. Maximum time from boning of pork loins until:

(a) The finished fabricated boneless pork loin roasts (type I), are trimmed, packed and placed in the freezer. 4 hours

(b) The boneless pork loins for slices (type II), are trimmed, formed, wrapped and placed in a freezer for further processing. 4 hours

(c) The pork slices (type II) are packaged and packed. 15 days

1/ Those boneless loins offered in a temperature range of 28°F to 46°F (-2°C to 8°C) inclusive shall not exceed 48°F (9°C) after trimming until the roasts and slices are in the freezing process.

3.2.1 Additional process requirements for boneless loins to be formed for fabrication of pork slices (type II).

1. Shall be prepared from boneless pork loins which are prepared to comply with the requirements of 3.3.1. Exception: Instead of netting, boneless loins shall be formed either by (1) stuffing into a 0.004-inch (0.102 mm) thick polyethylene bag that is mechanically closed to provide a snug fit or (2) by wrapping the loin in 0.001-inch (0.025 mm) thick cast polyvinyl chloride and forming in molds to meet the shape, thickness, and weight requirements of 3.3.2.

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The polyvinyl chloride or polyethylene bag shall conform to the pork loins to prevent freezer-burn and muscle distortion. The packaged boneless pork loins shall be frozen to an internal temperature of 28°F (-2°C) or below within 24 hours from the time they enter the freezer.

2. Prior to slicing, the product may be tempered to a temperature not to exceed 28°F (-2°C) to facilitate slicing. The tempering process shall exclude the use of steam or water. (Note: The forming, freezing, tempering, slicing operation shall be performed in sequence to maintain lot identity.) At no time shall the product be frozen and tempered more than once and the tempered product shall not be trimmed of softened (thawed) areas to comply with the 1/4 inch (6 mm) tolerance for this condition.

3. The formed pork loins shall be sliced in equipment designed to provide uniform weights and thickness as indicated by finished product requirements. Not more than 1 hour shall elapse from the time a formed pork loin is withdrawn from the freezer or tempering box until it is sliced.

4. The pork slices shall be packed and returned to the freezer within 2 hours after being sliced. In no case, during this time, shall the slices show evidence of soft (unfrozen) areas on any cut surface more than 1/4 inch (6 mm) inward from the edge.

3.2.2 Additional process requirements for pork loins to be used for country style ribs. A portion of the shoulder (blade) end of a loin may be used as raw material for the production of country style ribs in accordance with the requirements of PP-P-571. In this case, a contract for country style ribs under PPP-P-571 requirements shall run concurrently with a contract for pork loin roasts and slices, boneless, frozen, under the requirements of this document. The first three ribs of the shoulder end of the pork loin shall be utilized for country style ribs and the remainder of the pork loin shall be utilized for roasts and slices.

3.3 Finished product.

3.3.1 Type I (roasts) and boneless, trimmed, formed pork loins for fabricating type II (slices). The finished boneless pork roasts (type I) and boneless, trimmed, formed pork loins for fabricating pork slices (type II) shall comply with the following requirements:

1. Shall be free from foreign material (see table IV).

2. Shall be in excellent condition (see 6.3).

3. The following shall be removed and excluded from delivery:

a. Flank meat, tenderloin, tail muscle (longissimus costarum), false lean, blade meat (meat lying over the blade bone). 1/

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b. Bone, cartilage, blood clots, bruises (see 6.3).

c. Dehydrated tissue in excess of 1 inch (25 mm) square (see 6.3).

4. Shall show no visual evidence of freezer-burn.

5. Shall be cylindrical (may be slightly oval in cross-section) in shape as applicable, for type I (roasts) or trimmed, formed loins for fabricating type II (slices) (see 6.2.1 and 6.2.2).

6. Shall consist only of one intact loin (ham) half and one intact rib (blade) half, superimposed over the other so that the ham end of half of the loin and blade end of the other half are at opposite ends of the finished roast or trimmed, formed boneless loin.

7. Each end of the loin shall not extend more than 1 inch (25 mm) beyond the corresponding end of the other half of the loin (applicable to roasts (type I) only).

8. Shall be free from surface fat in excess of 1/2 inch (13 mm) average thickness, with no point exceeding 3/4 inch (19 mm) (applicable to roasts (type I) only).

9. The fat shall not extend in excess of 1/2 inch (13 mm) beyond the ventral edge of the major loin muscle.

10. Shall have the fat surface of the loin on the outside.

11. Shall have a suitable binding agent (see 3.1.2) applied to the interior lean surfaces of each half of the loin.

12. Shall be free from deep cuts and tears (see 6.3).

13. Shall be completely inclosed in elasticized netting material meeting the requirements of MIL-C-43619 (applicable to roasts (type I) only (see 6.2.1)).

14. Shall weigh not less than 3-1/2 pounds (1.59 kg) (applicable to roasts (type I)) only.

1/ The tail muscle (longissimus costarum) shall be removed by a cut through the natural seam to the point of its termination and then continue in a straight line parallel to the back line to the anterior end of the loin.

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3.3.2 Type II (pork slices). In addition to the requirements of 3.3.1 (applicable to boneless, trimmed, formed loins), boneless, fabricated, pork slices shall comply with the following requirements:

1. Each slice shall contain two full-cut boneless loin portions that are intact. Exception: Facing (end) slices need not contain two full-cut loin portions but they shall not consist of only one portion of meat or more than two portions of meat (see 6.3).

2. Shall weigh 5 ounces (142 g) ± 1/2 ounce (14 g). Facing (end) slices shall weigh not less than 4 ounces (113 g) nor more than 6 ounces (170 g) (see 6.3).

3. Shall be not less than 1/2 inch (13 mm) nor more than 7/8 inch (22 mm) in thickness. Facing (end) slices from the ends shall be not less than 3/8 inch (10 mm) nor more than 7/8 inch (22 mm) in thickness.

4. Shall be cylindrical in shape (may be slightly oval in cross-section).

5. Shall show at least 75 percent lean on each side of the two major cut surfaces formed by the slicing operation.

6. Each slice (except facing (end) slices), shall be cut at a right angle to the grain of the meat.

7. Meat dust shall not exceed an aggregate of a 1/2 inch (13 mm) square area and be more than 1/8 inch (3 mm) in thickness.

8. Shall not be cracked, fractured in part, or partially separated (hinged) (see 6.3).

9. Shall show no evidence of thawing and refreezing.

10. Each box shall contain not more than 10 percent by weight of facing (end) slices.

11. Shall be free of surface fat in excess of 1/2 inch (13 mm) average thickness with no point exceeding 3/4 inch (19 mm).

3.4 Freezing and frozen storage.

3.4.1 Freezing. The product shall be frozen utilizing one of the following processes.

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3.4.1.1 Conventional freezing process.

(1) The product properly packed, shall be blast frozen or frozen by equivalent means to provide forced air circulation while maintaining a freezer temperature of -10°F (-23°C) or less during the freezing process.

(2) The shipping containers shall be arranged so that the container surfaces are not in direct contact with each other for at least 24 hours after the start of the freezing process before restacking in a more convenient arrangement.

(3) The product shall remain in the freezing process until the temperature in the thickest part of the product reaches a temperature of 0°F (-18°C) or less.

3.4.1.2 Individual quick freezing process (IQF).

(1) Prior to packing, the product shall be frozen in cryogenic freezing equipment (gaseous nitrogen tunnel, rotating drum, freezer or other type of "quick freeze" equipment) to reduce the temperature to 0°F (-18°C) or less in the thickest part of the product within 1 hour after the product enters the freezer.

(2) Following the freezing process, the product shall be maintained at a uniform temperature of 0°F (-18°C) or less while being packed and prepared for delivery.

3.4.2 Frozen storage. Product handling subsequent to freezing shall be as follows:

(1) After being frozen and until time for delivery, the product shall be held at a uniform temperature of 0°F (-18°C) or less. The freezer temperature may fluctuate (not to exceed 6°F (-14°C)) for short periods to allow for entering the freezer to restock, withdraw, etc.

(2) The internal temperature in the thickest part of the product shall not exceed 0°F (-18°C) at time of shipment and delivery.

(3) The product shall be in excellent condition (see 6.3) and shall show no evidence of thawing, or refreezing when examined at destination.

3.5 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

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3.6 Plant qualification. The product shall originate, be processed, marked and handled in plants which are operated under the Meat and Poultry Inspection Regulations of the US Department of Agriculture. Finished product refrigeration facilities (freezer, storage and shipping) not operating under these regulations shall meet the requirements of MIL-STD-668.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified (see 6.1) in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Plant qualification conditions. The product furnished under this document shall be unacceptable if not produced and stored in plants which currently meet the qualification conditions of 3.6.

4.3 Quality conformance inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

4.3.1.1 Examination of bone-in pork loins prior to processing. Prior to Government verification inspection, the contractor shall assemble the bone-in pork loins into lots. Each lot shall be identified in a manner approved by the supervisory Government inspector. The contractor shall prepare, sign and furnish the Government Quality Assurance Representative (AQR) a report verifying the total weight, temperature, weight range, compliance with cutting requirements of 3.1.1 and selection requirements of 3.1.1.1 and condition of the lot. Failure of the contractor to assemble, identify, maintain lot identity, and furnish the signed report, shall be cause for rejection of the lot. Determination of compliance to the requirements of 3.1.1 and 3.1.1.1 shall be ascertained by the QAR. Failure of any bone-in loin to comply with one or more of the referenced requirements shall be cause for rejection of the lot. When foreign materials are found during the examination of the pork loins, the lot shall be referred to the US Department of Agriculture Inspector for disposition in compliance with inspection criteria of the Meat and Poultry Inspection Regulations of the US Department of Agriculture.

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4.3.1.2 Examination of binding agents. Examination of binding agents shall be made to determine compliance with the identity requirements of 3.1.2. Determination of compliance to the identity requirements shall be ascertained by examination of labels, invoices or other valid documents. In addition, materials shall be examined to determine compliance with the condition requirements of 3.1.2. The sample for examination shall be a 1 pound (0.45 kg) composite derived equally from five primary containers or all the containers if less than five form a lot. Nonconformance to one or more of the above referenced requirements shall be cause for rejection of the component lot or the finished product made therefrom, as applicable. When foreign materials are found during the examination, the lot shall be referred to the US Department of Agriculture Inspector for disposition in compliance with inspection criteria of the Meat and Poultry Inspection Regulations of the US Department of Agriculture.

4.3.1.3 Examination of elasticized netting material. Examination of elasticized netting material shall be in accordance with examination criteria of MIL-C-43619.

4.3.2 Testing of components. Testing of components described in this specification and changes in subsidiary specifications shall be as shown in table I. Test requirements shall be lot average requirements. Nonconformance to one or more test requirements shall be cause for rejection of the lot.

TABLE I. Testing of components 1/

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported to nearest	Test ref. and rqmt para
Polyethylene bag or sheet as applicable	One bag or sheet	Number of bags or sheets	S-1	Thickness of polyehtylene	0.0001 inch (0.001 mm)	4.4.1 5.1.1.2
Paperboard separator	One separator	Number of separators	S-1	Thickness of separators	0.001 inch (0.01 mm)	5.1.1.2
Polyethylene bag	One bag	Bags	S-1	Thickness	0.001 inch (0.01 mm)	4.4.1 3.2.1

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TABLE I. Testing of components 1/ (cont'd)

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported to nearest	Test ref. and rqmt para
Polyvinyl chloride	One sheet	Sheets	S-1	Thickness	0.0005 inch (0.001 mm)	4.4.1 3.2.1

1/ In lieu of testing for thickness, ascertainment of compliance with the requirements shall be determined by examination of labels, invoices or other valid documents. Nonconformance to requirements shall be cause for rejection of the lot.

4.3.3 In-process examination. Examination shall be made during processing to determine compliance with the requirements of 3.2, 3.2.1 and 3.2.2, as applicable. Records of processing times and temperatures shall be maintained. Nonconformance to one or more requirement(s), determined by actual examination or records, shall be cause for rejection of the lot. In addition, examination of boneless, trimmed, formed loins for fabricating type II (slices) shall be performed in accordance with applicable defects of table IV.

4.3.4 Examination of end item. Classification of defects found during examination shall be in accordance with tables III through VII and 4.3.4.1 through 4.3.4.5. The examination criteria shall be as shown in table II. AQLs shall be expressed as defects per hundred units.

TABLE II. End item examination criteria

Table	Inspection level	Sample unit	Lot size expressed in	AQLs Major A	Major B	Minor
III	I	One roast	Estimated number of roasts	-	-	4.0
IV	S-4	One roast or one boneless, trimmed, formed loin, as applicable	Estimated number of roasts or boneless, trimmed, formed loin as applicable	1/	2.5	6.5
V	I	One slice	Estimated no. of slices	1/	2.5	6.5

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TABLE II. End item examination criteria (cont'd)

Table	Inspection level	Sample unit	Lot size expressed in	AQLs Major A	Major B	Minor
VI	S-3	Contents of one shipping container	Number of shipping containers	-	1.5	-
VII	S-3	One filled and closed shipping container	Containers	-	2.5	6.5

1/ The finding of any Major A defect shall be cause for rejection of the lot.

TABLE III. Examination of net weight (type I only) 1/

Category	Defect
Minor	
201	Weight less than 3-1/2 pounds (1.59 kg).

1/ Weigh to a tolerance of 1/4 pound (0.11 kg).

TABLE IV. Examination of boneless pork roast (type I) and boneless, trimmed, formed loin for fabricating pork slices (type II) prior to freezing

Category			Defect
Major A	Major B	Minor	
101			Presence of foreign material. 1/
102			Not in excellent condition (see 6.3).
	151		Not netted (applicable to roasts (type I) only).
	152		Not compact, intact or firmly secured, i.e., not completely netted (applicable to roasts (type I) only).
	153		Suitable binding agent not applied to the interior lean surface of each half of the roast.
		201	Not cylindrical in shape (see 3.3.1, item 5).
	154		Presence of deep cut or tear (see 6.3).

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TABLE IV. Examination of boneless pork roast (type I) and boneless, trimmed, formed loin for fabricating pork slices (type II) prior to freezing (cont'd)

Category			Defect
Major A	Major B	Minor	
	155		Fat surface of each half of the loin not on the outside.
	156		Does not consist of one intact loin half and one intact rib half of the loin.
		202	End of one half extends more than 1 inch (25 mm) beyond either end of the adjacent half (applicable to roasts (type I) only.
		203	Ham end of loin half and blade end of rib half of the loin not at opposite ends of the roast.
	157		Presence of flank meat.
		204	Presence of tenderloin in excess of 1/2 inch (13 mm) in any dimension. 2/
	158		Presence of false lean.
	159		Presence of blade meat (meat lying over the blade bone).
	160		Presence of dehydrated tissue in excess of 1 inch (25 mm) square (see 6.3).
	161		Surface fat exceeds 1/2 inch (13 mm) average thickness (applicable to roasts (type I) only (see 6.3).
	162		Fat extends beyond the ventral edge of the major loin muscle more than 1/2 inch (13 mm) (applicable to roasts (type I). 3/
	163		Surface fat exceeds 3/4 inch (19 mm) at any point (applicable to roasts (type I) only).
		205	Presence of longissimus costarum muscle greater than 1-1/2 inches (38 mm) in width or length.
		206	Presence of bone and cartilage (see 6.3). 2/
		207	Presence of blood clot(s) and bruise(s) (see 6.3). 2/
		208	The thin meat at the anterior end not removed by a straight cut parallel to the backline that begins at the termination of the tail muscle (longissimus costarum).

1/ Within the total quantity represented by all sample units of the sample size specified, defects equal to or in excess of the foreign material tolerances specified in 6.3 shall be cause for rejection of the inspection lot.

2/ Measure to a tolerance of 1/8 inch (3 mm).

3/ Measure to a tolerance of 1/4 inch (6 mm).

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TABLE V. Examination of boneless pork slices (type II)

Category			Defect
Major A	Major B	Minor	
	151		Presence of soft (nonfrozen) areas (see 3.2.1).
	152		Evidence of thawing and refreezing.
101			Presence of foreign material. 1/
102			Not in excellent condition (see 6.3).
		201	Not cylindrical in shape (see 3.3.2, item 4).
	153		Not two full-cut intact slices (except for facing slices).
	154		Cracked, fractured in part, or partially separated (hinged) (see 6.3).
		202	Weight of other than facing slices: More than 5-1/2 ounces (156 g) or less than 4-1/2 ounces (128 g). 2/
	155		Weight of facing slices: More than 6 ounces (170 g) or less than 4 ounces (113 g). 3/
		203	Thickness of other than facing slices: More than 7/8 inch (22 mm) or less than 1/2 inch (13 mm). 4/
	156		Thickness of facing slices: More than 7/8 inch (22 mm) or less than 3/8 inch (10 mm). 4/
	157		Not cut at right angles to the grain of the meat (except facing slices).
	158		Surface fat exceeds 1/2 inch (13 mm) average thickness (see 6.3).
	159		Surface fat exceeds 3/4 inch (19 mm) at any point.
	160		Presence of facing slice from ends consisting of only one portion of meat.
	161		Presence of facing slice made of more than two portions of meat.
		204	Meat dust having an aggregate area of more than 1/2 inch (13 mm) square 5/, and 1/8 inch (3 mm) in thickness. 4/
		205	Less than 75 percent lean tissue on each side of the two major surfaces. 6/
		206	Presence of dehydration.

1/ Within the total quantity represented by all sample units of the sample size specified, defects equal to or in excess of the foreign material tolerances specified in 6.3 shall be cause for rejection of the inspection lot.

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2/ Weigh to a tolerance of 1/4 ounce (7 g).

3/ Weigh to a tolerance of 1/2 ounce (14 g).

4/ Measure to a tolerance of 1/16 inch (2 mm).

5/ Measure to a tolerance of 1/4 inch (6 mm) square.

6/ Measure to a tolerance of 5 percent.

TABLE VI. Examination for percent of facing slices

Category	Defect
Major B	
151	Presence of more than 10 percent by weight of facing slices in shipping container. 1/

1/ Measure to the nearest whole percent.

TABLE VII. Examination for packaging of slices 1/

Category		Defect
Major B	Minor	
151		Each layer not separated by a separator that prevents slices from adhering to each other.
152		Unclean packaging material.
	201	Shipping container not lined as specified.
	202	Bag not of sufficient size.
	203	Meat slices not arranged as specified.
	204	Separators not material specified.
	205	Separator dimensions not as specified.
	206	Wax on paperboard missing or insufficient to show presence of wax on the fingernail when scrapped lightly across the board.
	207	Separators not placed as specified.
	208	Bag not folded down over the top of the meat.

1/ See 5.1.1.2 or 5.1.2, as applicable.

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4.3.4.1 Examination for packaging of roast. Examination of level B and level C packaging shall be in accordance with the end item inspection criteria of MIL-M-1018, except that for level C pack the physical requirements of the packaging materials are not applicable.

4.3.4.2 Examination of shipping container. Examination of the filled and closed shipping containers shall be in accordance with the appendix to PPP-B-636 or PPP-B-1163, as applicable. In addition, the following defects shall be included in the table of examination. Major: (1) NSN, contract No., date of pack or item description markings; missing, incorrect, or illegible. (2) Imparts odor or color to product. (3) Color of reinforcing bands not as specified. (4) Not snug fitting (applicable to type II only). Minor: (1) More than 55 pounds (24.95 kg) of product as indicated by marked net weight. (2) Not snug fitting (applicable to type I only). (3) Other marking requirements not as specified. For level C pack, examination shall be in accordance with aforementioned specifications, except that only the additional defects, as applicable, and closure requirements (see 5.2.2) shall apply.

4.3.4.3 Examination of shipping container for net weight. The sample unit shall be one filled shipping container. Lot size shall be expressed in terms of the number of sample units. Sample size shall be the number of sample units indicated by inspection level S-4. The lot shall be rejected if, by using recognized statistical methods, the sample average net weight is determined to be less than the sample average marked net weight.

4.3.4.4 Examination at time of freezing and during frozen storage. Examination shall be made at time of freezing and during frozen storage to determine compliance with the applicable requirements of 3.4, as applicable. Records of time and temperature shall be maintained. Nonconformance to one or more of the above referenced requirements reflected by actual examination or by records shall be cause for rejection of the lot. In addition, evidence of thawing or refreezing shall also be cause for rejection of the lot.

4.3.4.5 Examination at time of shipment and delivery. Examination shall be made at time of shipment and delivery to determine compliance with the applicable requirements of 3.4, as applicable. The sample unit shall be one filled and closed shipping container. Lot size shall be expressed in terms of the sample unit and the sample size shall be the number of sample units indicated by inspection level S-2. Finding of any temperature not in accordance with specification requirements shall be cause for rejection of the lot. In addition, evidence of thawing or refreezing, or product not in excellent condition at destination shall be cause for rejection of the lot.

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5. PACKAGING

5.1 Preservation-packaging. Packaging shall be level B or C, as specified (see 6.1).

5.1.1 Level B.

5.1.1.1 Roasts. The finished roasts, specified in 3.3.1, shall be packaged in accordance with the level B packaging requirements for pork loin, boned, of MIL-M-1018.

5.1.1.2 Slices. The slices, specified in 3.3.2 shall be packaged into a box as specified in 5.2. The box shall be lined with a bag of sufficient size to completely line the interior surfaces of the box, and shall be so constructed that the opposing lips of the bag have an overlap of not less than 5 inches (127 mm) along the center of the longer axis after the bag is folded down on the packed meat. The bag shall be 0.0015-inch (0.0381 mm) thick polyethylene in accordance with L-P-378. The slices shall be arranged in layers with no slice overlapping another slice. The layers shall be separated by a sheet of not less than 0.015-inch (0.381 mm) solid manila, kraft, or sulphite paperboard not more than 1/8 inch (3 mm) shorter in dimension than the length and width of the box to prevent slices from adhering to each other. The paperboard shall be coated overall with: (1) wax, sufficient to show the presence of wax on the fingernail when it has been scraped lightly across the board; or (2) a minimum of 0.0005-inch (0.0127 mm) thick polyethylene. Alternatively, the separator shall be a sheet of 0.0015-inch (0.0381 mm) thick polyethylene in accordance with L-P-378. The sheet shall be not less than 1 inch (25 mm) longer than the length and width of the container. In addition, a separator as described above, shall be placed between the bottom of the bag and the first layer of the meat, and one shall also be placed on the surface of the top layer. The bag shall be neatly folded down over the top of the meat and the box closed.

5.1.2 Level C. The product shall be packaged in accordance with 5.1.1, except that the physical requirements of the packaging materials are not applicable. The types of material and style of packaging referenced shall be used.

5.2 Packing. Packing shall be level B or C, as specified (see 6.1).

5.2.1 Level B packing. Not more than 55 pounds (24.95 kg) of roasts or slices, as applicable, packaged as specified in 5.1, shall be packed in a snug-fitting (see 6.5) fiberboard shipping container constructed in accordance with style FTC, V3c, V3s or V4s of PPP-B-636. The container shall impart no odor or color to the product. Alternatively, the shipping container shall be a wax-impregnated fiberboard box constructed in accordance with style RSC, class I, type SWCFI, grade 275 of PPP-B-1163. Closure of the wax-impregnated shipping container shall be with hot melt adhesive in accordance with the appendix of PPP-B-636. All shipping containers shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636. Reinforcing bands shall be uncolored or of a color which contrasts with black. Wax-impregnated boxes shall be reinforced with two lengthwise bands and one girthwise band in accordance with the appendix of PPP-B-636.

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5.2.1.1 Alternate shipping container. When specified (see 6.1 and 6.6), the shipping container shall be a snug-fitting fiberboard box, constructed and reinforced in accordance with style FTC, W5c or W5s of PPP-B-636 except that two reinforcing bands may be used, one lengthwise and one girthwise. Alternatively, the shipping container shall be a wax-impregnated fiberboard box constructed in accordance with style RSC, class I, type SWCFI, grade 200 of PPP-B-1163. Closure of the wax-impregnated shipping container shall be with hot melt adhesive in accordance with the appendix of PPP-B-636. All shipping containers shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636. Reinforcing bands shall be uncolored or of a color which contrasts with black. Wax-impregnated boxes shall be reinforced with two lengthwise bands and one girthwise band in accordance with the appendix of PPP-B-636.

5.2.2 Level C packing. The product, packaged as specified in 5.1, shall be packed in shipping containers complying with Uniform Freight Classification or National Motor Freight Classification, as applicable, except that closure of fiberboard boxes shall be in accordance with the appendix of PPP-B-636.

5.3 Marking.

5.3.1 Packages. Labeling shall conform to Meat and Poultry Inspection Regulations of the US Department of Agriculture.

5.3.2 Shipping containers. Shipping containers shall be marked in accordance with MIL-STD-129 and as specified in 5.3.2.1.

5.3.2.1 Additional marking. On top of each container, the following information shall be printed in bold capital letters between 1 inch (25 mm) and 1-1/2 inches (38 mm) high, all letters being of the same height:

PERISHABLE - KEEP FROZEN

(0°F (-18°C) or below)

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of product required (see 1.2).
(c) Level of packaging required (see 5.1).
(d) Level of packing required (see 5.2).
(e) When packing specified in 5.2.1.1 is required (see 6.6).
(f) **When other than the contractor is to perform the examinations and tests (see 4.1).**

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6.2 Suggested operating procedures. As a suggestion to the manufacturer, not as a specification requirement, the procedures outlined below will normally produce a product meeting the requirements of this specification.

6.2.1 Roasts. The trimmed boneless loin should be cut in half, perpendicular to the length of the loin, so that the loin (hip) section and blade (rib) section are the same length. The two sections of the same loin, one loin half and one rib half, should be reversed and placed together, lean to lean, so that the loin end of one section and the blade end of one section, are at opposite ends of the roast. The roast should have a uniform diameter for its entire length. A binding agent of the quality specified in 3.1.2 should be applied uniformly on the lean interior surface of each loin half, prior to placing the rib and loin halves together. After the binding agent is added, the two halves should be immediately completely enclosed in elastic netting.

6.2.2 Slices. The slices should be cut from loins as prepared in 6.2.1, except that the halves should not be netted and should be formed in casings or cylindrical rigid molds prior to freezing and slicing. In order to utilize the ends of the loins for slices, the ends should be perpendicular to the length of the loin. This may be accomplished with disks positioned in the ends of the casings or molds. Care should be taken in selecting casing sizes which will yield formed slices complying with both thickness and weight requirements of this specification. Product in casings should be overwrapped during freezing to prevent freezer-burn.

6.3 Definitions of terms. For purpose of this specification the following terms shall apply.

6.3.1 Surface fat. Surface fat is fat that is on the outer covering of meat or appears at the surface of a roast or outer edge of a slice after fabrication. Seam fat which appears on the surface of a roast shall comply with the surface fat requirement, if the surface of the lean adjacent to the seam is covered with not more than 1/2 inch (13 mm) of fat and the exposed surface of the seam fat follows the contour of the surface in that area. In the case of slices, when depressions occur in the lean to form a pocket of fat, or when seam fat appears at the outer edge of a slice, a straight edge shall be used to span the gap, and that fat which extends beyond the straight edge shall be considered surface fat. The fat span (bridge) shall be not more than 1 inch (25 mm) in width.

6.3.2 Bruise. A bruise is a damaged portion of muscle or fatty tissue, as evidenced by opaqueness of connective tissue or darkened flesh which exceeds 1 inch (25 mm) in any dimension.

6.3.3 Blood clot. A blood clot is an organized, coagulated, opaque mass of blood on the surface of fat or muscle fibers which exceeds 1/2 inch (13 mm) in any dimension.

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6.3.4 Deep cut or tear. A deep cut or tear is one that penetrates the lean to a depth of more than 1 inch (25 mm) and the aggregate of such cuts exceeds one-half the length of the roast.

6.3.5 Bone. Bone is connective tissue whose matrix is hardened by various salts of calcium and potassium. Functional definition: Bone is ossified tissue greater than 1/4 inch (6 mm) in any dimension. Ossified tissue 1/4 inch (6 mm) or less shall not be classified as bone.

6.3.6 Cartilage. Cartilage is elastic, unossified bone tissue greater than 1/4 inch (6 mm) in any dimension. Unossified material 1/4 inch (6 mm) or less shall not be classified as cartilage.

6.3.7 Cracked slice. A slice shall be considered cracked (fractured) when two or more separate or hinged pieces of meat result from that slice (examine in a frozen state). Seam separation shall not be considered a defect unless complete separation into two or more distinct pieces occur.

6.3.8 Facing (end) slice. The facing or end slice is the first slice removed from either end of the formed roast.

6.3.9 Excellent condition. Meat in excellent condition has a degree of freshness corresponding to that exhibited by bone-in meat which is promptly chilled after slaughter to an internal temperature of 28° to 40°F (-2° to 4°C) and maintained within this range with minimal fluctuation and with proper air circulation for a period not exceeding ten days. Meat packaged by heat-shrink vacummized methods may develop a smothered beef odor and/or a darkening of exposed muscle tissue. Fresh bone-in meat may develop surface dehydration and/or a darkening of exposed muscle tissue. Meat displaying any of the undesirable characteristics mentioned above will retain an excellent condition classification only if it can be processed for use without excessive trimming. Meat which has developed a stickiness, sliminess, or discoloration in conjunction with an off-odor shall not be considered in excellent condition. (This definition has been provided by the Department of the Army, Office of the Surgeon General, Washington, DC).

6.3.10 Foreign material. Foreign material is any extraneous matter which does not organically belong where found, which has been introduced from the outside, or which does not naturally occur in the quantity found at the location examined.

a. Particles.

(1) One or more individual particles, including dust, paint chips, sawdust, and wood splinters in excess of 1/8 inch (3 mm) in greatest dimension;

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(2) Smaller particles that in an aggregate cover an area of more than 1/2 inch (13 mm) in diameter;

(3) One or more individual particles which are visually determined to be metal, glass, or hard plastic.

b. Soft extraneous material (string, plastic wrap or tape, paper, cardboard, etc.) one or more individual pieces more than 1/2 inch (13 mm) in greatest dimension.

c. Insects. One or more insects, except that one single gnat or fruit fly is considered insignificant and is acceptable.

d. Skin (with or without hair or visible hair roots) individually or in an aggregate in excess of the area contained in a 1 inch (25 mm) square.

e. Hair and/or visible hair roots in excess of three single strands of hair or five visible hair roots or one clump of hairs or visible hair roots (strands too numerous to count) in one area.

f. Lung tissue. Any amount.

g. Stains. One or more grease stains or discoloration that individually or in aggregate covers an area more than that of a circle 1/2 inch (13 mm) in diameter.

h. Ingesta. Any amount.

i. Fecal amount. Any positive identified amount.

When appropriate the final decision with respect to the interpretation of the sanitary, wholesome or esthetic significance of foreign material defects in end item shall remain with a military veterinary officer.

(This definition has been provided by the Department of the Army, Health Services Command, Fort Sam Houston, TX 78234.)

6.4 Appropriate level of pack. Based on conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.14D/DSAR 4145.7.

6.5 Snug-fitting. The fitting of product in its container in a manner which prevents the movement or shifting of the product during subsequent shipment and handling.

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6.6 Intended use of alternate shipping container. Packing specified in 5.2.1.1 is intended for use when knowledge of the destination or mode of transportation indicates that a lower grade of board may be used.

6.7 Plant sanitary conditions. Award of contract for the product specified in this document will be limited to plants known to maintain the required sanitation conditions of 3.6 (also see 4.2).

6.8 Metric equivalents. Metric equivalents, indicated in parentheses throughout this document, are based on practices, conversion factors, and symbols specified in ASTM E 380 Standard for Metric Practice, and are for information only. In each instance, the value stated in US customary units shall be controlling.

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD
Navy - MC, MS
DP - SS

Preparing activity:

Army - GL

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-P-35098F Pork Loin Roasts & Slices, Boneless, Frozen

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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